The leximin core selection on exact games

Draft fragment: necessity theorem, interval lemma, and the open crux

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Status: mathematical core of a paper in progress. Contains one unconditional theorem (exactness is necessary), one proven structural lemma, a conditional main theorem, and two precisely stated open lemmas with computational evidence. Bibliography keys refer to the companion papers' reference lists.

1 The leximin-above-floors core selection

Let (N, v) be a balanced TU game, $M = v(N)$, and floors $m_i := v(\{i\})$. For $x \in \mathbb{R}^N$ write the *gains* $g_i(x) = x_i - m_i$. The *leximin-above-floors* selection is

$$\phi(v) := \arg \operatorname*{leximin}_{x \in \mathrm{Core}(v)} \big(g_1(x), \ldots, g_n(x)\big),$$

the unique core allocation whose ascending-sorted gain vector is lexicographically largest. It is single-valued (leximin over a polytope) and is computed by the Maschler-type scheme over player coordinates: repeatedly raise a common lower level s of the free players' gains as far as the core allows, fix ("pin") every free player whose gain cannot exceed $m_i + s$ on the level-optimal face, and recurse on the rest. On *exact* games $m_i = \min_{\mathrm{Core}} x_i$, so $g_i \geq 0$ and ϕ egalitarianly distributes the surplus $x_i - v(\{i\})$ inside the core; on convex games it is the constrained egalitarian solution of Dutta and Ray (1989). Implementation (exact rational + float) in `src/leximin.py`, validated against those two coincidences.

We ask whether ϕ is *coalitionally monotonic*: if $v' = v + d\,\mathbf{1}_S$ for a single coalition S and $d > 0$, is $\phi_i(v') \geq \phi_i(v)$ for every $i \in S$? A positive answer on exact games would exhibit an explicit coalitionally monotonic core selection on that class.

2 Exactness is necessary

Coalitional monotonicity of ϕ fails on balanced games in general—already at $n = 4$, and even at $n = 5$ where every floor is core-attained. So exactness cannot be weakened to balancedness.

Theorem 2.1. *There is a balanced (non-exact) five-player game v, a coalition S with $|S| = 3$, and $d > 0$ with $v + d\,\mathbf{1}_S$ balanced, such that some $i \in S$ has $\phi_i(v + d\,\mathbf{1}_S) < \phi_i(v)$. The same holds for a four-player game.*

Example 2.2 ($n = 5$; floors attained)**.** *Let $N = \{1, \dots, 5\}$ with $v(N) = 7$ and (all other coalitions as listed in the supplement) in particular*

$$v(\{1,2,3\}) = 3, \quad v(\{2,3\}) = 3, \quad v(\{1,3,5\}) = v(\{3,5\}) = 4, \quad v(\{1,5\}) = -1.$$

Floors are $m = (0,1,1,1,2)$, and $m_i = \min_{\mathrm{Core}} x_i$ for every *i (all floors attained); the game is balanced but not exact (e.g. $\min_{\mathrm{Core}} x(\{1,5\}) = 1 > -1$). Then*

$$\phi(v) = \left(\tfrac{1}{2}, \tfrac{3}{2}, \tfrac{3}{2}, 1, \tfrac{5}{2}\right), \qquad \phi(v + \mathbf{1}_{\{1,2,3\}}) = (1,1,2,1,2),$$

both in their respective cores, and player $2 \in \{1,2,3\}$ drops *from $\frac{3}{2}$ to 1 when $v(\{1,2,3\})$ is raised from 3 to 4. Thus exactness at the singletons (floor attainment) does not suffice: it is exactness at higher coalitions that the positive theorem needs.*

Example 2.3 ($n = 4$; floor unattained)**.** *Let $N = \{1,2,3,4\}$, $v(N) = 20$, with $v(\{1\}) = -3$, $v(\{1,2\}) = 11$, $v(\{1,3\}) = 7$, $v(\{2,3\}) = 4$, $v(\{1,2,3\}) = 13$, $v(\{2,4\}) = 10$, $v(\{1,3,4\}) = 9$, $v(\{2,3,4\}) = 13$, and the remaining values in the supplement. Here $v(\{1\}) = -3 < 1 = \min_{\mathrm{Core}} x_1$. Then $\phi(v) = (\frac{5}{2}, \frac{17}{2}, \frac{9}{2}, \frac{9}{2})$ and $\phi(v + 3\,\mathbf{1}_{\{1,2,3\}}) = (2,9,5,4)$: player $1 \in \{1,2,3\}$ drops from $\frac{5}{2}$ to 2.*

Both examples are certified in exact rational arithmetic (ϕ in the core on both sides, the drop exact); see the supplement and `src/hunt_leximin_nonexact.py`.

3 Reduction on exact games

Fix an exact v, a proper coalition S with $|S| \geq 2$, and $d > 0$ with $v' = v + d\,\mathbf{1}_S$ exact. Since $|S| \geq 2$ the floors are unchanged, and

$$\mathrm{Core}(v') = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\} \subseteq \mathrm{Core}(v).$$

For $\tau \in [v(S), v(S)+d]$ let v_τ agree with v except $v_\tau(S) = \tau$, so $\mathrm{Core}(v_\tau) = \mathrm{Core}(v) \cap \{x(S) \geq \tau\}$ interpolates $\mathrm{Core}(v)$ and $\mathrm{Core}(v')$.

Lemma 3.1 (exact interval)**.** *Every v_τ, $\tau \in [v(S), v(S)+d]$, is exact.*

Proof. $\mathrm{Core}(v_\tau) \supseteq \mathrm{Core}(v') \neq \emptyset$, so v_τ is balanced. Exactness needs $\min_{\mathrm{Core}(v_\tau)} x(T) = v_\tau(T)$ for all T. For $T = S$: $\min_{\mathrm{Core}(v_\tau)} x(S) = \max\{\tau, \min_{\mathrm{Core}(v)} x(S)\} = \max\{\tau, v(S)\} = \tau$. For $T \neq S$: put $f_T(\tau) = \min_{\mathrm{Core}(v_\tau)} x(T)$. As τ grows $\mathrm{Core}(v_\tau)$ shrinks, so f_T is nondecreasing; and $\mathrm{Core}(v_\tau) \subseteq \mathrm{Core}(v)$ gives $f_T(\tau) \geq \min_{\mathrm{Core}(v)} x(T) = v(T)$. At the endpoints $f_T = v(T)$ because v and v' are exact. A nondecreasing function equal to $v(T)$ at both endpoints and $\geq v(T)$ between is $\equiv v(T)$. Hence $f_T(\tau) = v(T) = v_\tau(T)$. □

So the bump is realized by a continuous path of *exact* games in which only the lower bound τ on $x(S)$ rises. This isolates the role of exactness: for a merely balanced v the interior of the path leaves exactness (Lemma 3.1 fails), and that is precisely the room inhabited by the counterexamples of Section 2. Coalitional monotonicity on exact games is therefore equivalent to the one-parameter statement:

Lemma 3.2 (monotone path — OPEN)**.** *Along the exact path v_τ, the map $\tau \mapsto \phi_i(v_\tau)$ is nondecreasing for every $i \in S$ (and, empirically, nonincreasing for $i \notin S$).*

Table 1: Computational evidence (float LP, exact-certifiable) for the reduction. $(\star_S)$ evaluated at ϕ of the base game; "binding" bumps only.

statement	sample	violations
ϕ coalitionally monotone on exact games	2000+ bumps, $n \le 6$	0
(D) exact $\Rightarrow (\star_S)$ at ϕ	39/39, 13/13	0
monotone $\Rightarrow (\star_S)$	all balanced	0
$(\star_S)$ along the path $\Rightarrow$ monotone	600	0
$(\star_S)$ at one endpoint $\Rightarrow$ monotone	1696 balanced	22 (all non-exact)

Two reductions cut the work down further. First, the common case is trivial:

Lemma 3.3 (non-binding)**.** *If $\phi(v)(S) \ge v(S) + d$ then $\phi(v') = \phi(v)$.*

Proof. $\phi(v)$ is the leximin point of $\mathrm{Core}(v) \supseteq \mathrm{Core}(v')$ and lies in $\mathrm{Core}(v')$, hence is the leximin point of the smaller set too; ϕ is single-valued. □

Binding ($\phi(v)(S) < v(S) + d$) requires $\phi(v)(S) < \max_{\mathrm{Core}} x(S)$ and is empirically rare. In every binding case observed (thousands of exact-preserving bumps at $n \le 6$, zero violations) one finds $\phi(v')(S) = v(S) + d$ and the two-sided pattern $\phi_i(v') \ge \phi_i(v)$ for $i \in S$, $\le$ for $i \notin S$.

A cone criterion. For $x = \phi(w)$ with tight set $\mathcal{T}(x) = \{T : x(T) = w(T)\}$, say $(\star_S)$ holds if there is $e \in \mathbb{R}^N$ with

$$e(N) = 0, \quad e(S) > 0, \quad e_i \ge 0 \ \ (i \in S), \quad e(T) \ge 0 \ \ (T \in \mathcal{T}(x)) :$$

a first-order direction raising $x(S)$ without lowering any member of S or breaking a tight core constraint. Empirically $(\star_S)$ is *necessary* for the bump to be monotone (it fails at $\phi(v)$ in Example 2.2, which is why player 2 must drop), and exactness supplies it. This suggests the skeleton

(A) Lemma 3.1: the bump path is exact;

(D) each exact v_τ satisfies $(\star_S)$ at $\phi(v_\tau)$;

(E′) $(\star_S)$ along the whole path implies Lemma 3.2,

whose composition is the theorem. The supporting evidence is summarized in Table 1; (D) and (E′) remain open, and $(\star_S)$ alone is *not* sufficient (the last row), so a proof of (E′) must use exactness along the whole path, not merely $(\star_S)$ at one endpoint.

The open problem, sharply. A Farkas dual turns (D) into a statement about a single maximal face of the core. For a core point y written as a mixture of core points, $y(T) \ge v(T) = \phi(T)$ at every tight T, so $e = y - \phi$ meets the tight-cone part of $(\star_S)$ automatically; hence

$$\begin{aligned}(\star_S) &\iff \exists\, y \in \mathrm{Core}(v) : \ y_i \ge \phi_i \ (i \in S) \text{ and } y(S) > \phi(S) \\ &\iff \max\{y(S) : y \in \mathrm{Core}(v),\ y_i \ge \phi_i \ (i \in S)\} > \phi(S).\end{aligned}$$

Dualizing (verified in 200/200 random balanced cases, exact-certifiable):

($\star_S$) *fails* at $\phi(v)$ if and only if $\phi(v)$ *maximizes some positive-S-weighted sum* $\sum_{k\in S} r_k\, x_k$ (with weights $r_k \geq 1$ for $k \in S$ and 0 otherwise) over $\mathrm{Core}(v)$.

Equivalently, the tight coalitions $\{T : \phi(T) = v(T)\}$ carry nonnegative weights and a $\beta \geq 1$ covering each $k \notin S$ exactly β and each $k \in S$ at most $\beta - 1$. Thus **Lemma 3.2's hypothesis (D) is exactly the assertion that an exact game's leximin point never lies on a positive-S-weighted maximal face of its core at a binding-bumpable S.** Uniform weights $r_k \equiv c$ are already excluded, since maximizing $c\,x(S)$ would force $\phi(S) = \max_{\mathrm{Core}} x(S)$, contrary to binding-bumpability; *the entire open content is that exactness excludes the non-uniform such faces.* The non-exact counterexamples of Section 2 realize exactly such faces—this is why ($\star_S$) fails there and a member of S drops. (A corner scan for a deficient S-member pinned at its core-maximum found no occurrence in any exact binding case, $0/16^{+}$.)

The natural certificate is provably wrong. Where ($\star_S$) *does* fail, one would like to certify that S is not exactly bumpable — $d^*(S) := \min_{C\neq S}\big(\max\{x(S) : x \in \mathrm{Core}(v),\, x(C) = v(C)\} - v(S)\big) = 0$ — directly from the failure certificate: take the cover $\{\lambda_U\}$ dual to the failure, $W = \bigcup\{U : \lambda_U > 0\}$, and propose $C = W \cap S$. This route is closed by a six-player exact game (supplement, `WITNESS_KEY_FALSE_n6`): ($\star_S$) fails at $S = \{1, 3, 4, 5\}$ with cover $\{2, 3\}, \{1, 2, 6\}, \{3, 4, 6\}$, yet a core point x with $x(W \cap S) = v(W \cap S)$ attains $x(S) = 24 > \phi(S) = 71/3$, so $W \cap S$ certifies nothing; the associated slack inequality $\phi(W) - v(W) \leq \phi(W \cap S) - v(W \cap S)$ fails by $1/3$. Such failures are rare ($\approx 0.1\%$ of ($\star_S$)-failures at $n \in \{6, 7\}$, found only under generators biased toward thin cores with nested tight chains), which is why sampling first suggested the cover certificate was universal. In every observed failure, however, the operative bound survives: there is a coalition C with $M^{(C)} := \max\{x(S) : x \in \mathrm{Core}(v),\, x(C) = v(C)\} \leq \phi(S)$, so that S cannot be bumped past $\phi(S)$ while preserving exactness (in a $\approx 0.2\%$ minority $d^*(S) > 0$ — exact six-player witnesses with $d^* = 1$ and $d^* = 2$ are included in the supplement — but always $d^*(S) \leq \phi(S) - v(S)$). Empirically the certificate has sharp structure: C can always be taken as a subset of S contained in a *single stratum* of the leximin computation — almost always a singleton — and it is provably never ϕ-tight (a ϕ-tight certificate would force $\phi(S) = v(S)$, contradicting positive slack). The open problem is thus sharpened: produce, from the structure of a ($\star_S$)-failure at the leximin point of an exact game, a single-stratum $C \subseteq S$ with $M^{(C)} \leq \phi(S)$. Empirically the certificate is pinned further: writing r for the residuals of the failure's dual cover ($r_i \geq 1$ on S), a *minimum-residual* member $k \in S$ (one with $r_k = 1$) certifies in 99.6% ($n = 6$) to 100% ($n = 7$) of observed failures, the residue requiring a same-stratum pair. By strong duality the singleton form is equivalent to exhibiting nonnegative weights on ϕ-tight coalitions with coverage β on $N \setminus S$, $\beta - 1$ on $S \setminus \{k\}$, and at least $\beta - 1$ on k; weights supported inside S suffice in only about two thirds of cases, so a proof must construct this dual cover from ϕ-tight coalitions *crossing* S. Equivalently, for a minimum-residual k the claim reduces to the single inequality $\sum_{i\in S\setminus\{k\}}(r_i - 1)\,(x_i - \phi_i) \leq 0$ for every core x with $x_k = v(\{k\})$: flooring the most-covered member must push the less-covered members of S, in the $(r_i - 1)$-weighting, to or above their leximin values. This inequality — which the residual weights alone do not force — is the precise frontier at which we leave the problem.

Remark 3.4. *Lemma 3.2 is a leximin parametric-monotonicity statement; its cousin the nucleolus is* not *coalitionally monotone, so exactness (Lemma 3.1) is doing real work and the "self-defeating within-S transfer" heuristic is not by itself a proof. We leave (D), (E′) open and record the conditional theorem:* on exact games, ϕ is coalitionally monotone provided Lemma 3.2 holds*; the necessity direction (Theorem 2.1) is unconditional.*

References

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